TaXer: Leveraging Feature Extraction for Long-term Forecasting of Strongly Periodic Sequences

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Abstract

While Transformer-based models have achieved remarkable success in long-term time series forecasting by capturing multivariate correlations, they often struggle to explicitly exploit the inherent multi-periodicity of real-world data. In this paper, we propose TaXer, a novel architecture designed to enhance forecasting accuracy by integrating deep periodic feature extraction with an inverted Transformer framework. TaXer employs a TimesNet-inspired mechanism as a periodic feature extractor to capture 2D temporal variations in the frequency domain, which are then utilized as global exogenous context. These features are cross-attentioned with patch-based endogenous representations, allowing the model to reconcile local fine-grained details with global periodic structures.

Furthermore, we introduce MeteoPolitan, a new periodicity-focused benchmark dataset , manually curated from global meteorological indicators (Berlin, Los Angeles, New York, Tokyo, and Seoul) and regional weather data from South Korea (Busan, Daegu, Gangneung, Gwangju and Seoul). Extensive experiments demonstrate that TaXer significantly outperforms current state-of-the-art models, including iTransformer and TimeXer, particularly on datasets with complex seasonal and periodic patterns. Our results suggest that explicitly modeling periodic variations as exogenous global tokens provides superior robustness in long-term horizons. The code and datasets are publicly available at https://github.com/Hyeonseop-Shin/DL-Project.

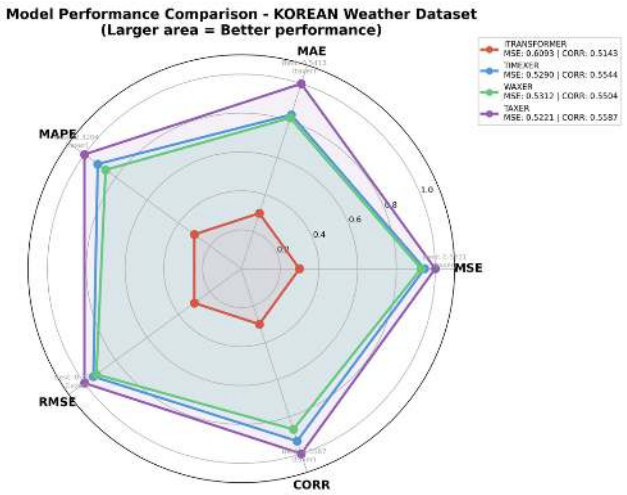


Figure 1. **Model performance Comparison for Korean dataset of *MeteoPolitan*.** Radar chart shows the errors(MSE, MAE, MAPE, RMSE, CORR) of each algorithms(iTransformer, TimeXer, WaXer, TaXer). TaXer shows outstanding result compared to other algorithms.

1. Introduction

Time-series forecasting is a fundamental task across various scientific and industrial domains, ranging from energy management to meteorological analysis. In particular, Long-Term Time-Series Forecasting (LTSF) presents a formidable challenge because prediction errors tend to accumulate as the forecasting horizon extends, often leading to performance degradation in existing deep models. While recent advancements in Transformer-based architectures have shown promise by capturing multivariate correlations, they frequently struggle to explicitly exploit the intrinsic characteristics of data that exhibit strong periodicity, such as atmospheric fluctuations. Accurate modeling of these periodic oscillations is essential for ensuring the reliability of predictions over extended horizons.

Existing approaches have attempted to model temporal dependencies using various mechanisms. Standard Transformers [12] typically apply attention over temporal tokens formed by multiple variates at a single timestamp, which can fuse distinct physical measurements and obscure variate-centric periodic structures. To overcome these limitations, recent studies such as TimesNet [14] have proposed transforming 1D time-series into 2D tensors to dis-

entangle multi-periodicity into intra-period and inter-period variations. Furthermore, TimeXer [13] introduced a method to incorporate exogenous variables through cross-attention to reconcile target series with external informative context. Despite these efforts, effectively capturing both fine-grained local context and broad global periodic trends simultaneously remains an open challenge in forecasting strongly periodic sequences.

In this paper, we propose TaXer, a novel hybrid architecture designed to maximize forecasting performance for data with dominant periodicity. TaXer synergistically integrates the strengths of inverted Transformer [6] and TimesNet's multi-periodic 2D variation modeling. Periodic features extracted via a TimesNet mechanism which serve as global exogenous tokens are cross-attentioned with patch-based endogenous representations, allowing the model to bridge causal information between broad periodic structures and local temporal patches.

A distinct contribution of this work is the establishment of *MeteoPolitan*, a high-fidelity periodic benchmark dataset. We curated meteorological data from global authorities, including the KMA[5], NOAA[8], JMA[3], and DWD[1], to address the quality limitations of standard public benchmarks. We implemented a systematic data engineering pipeline that utilizes the Auto-Correlation Function (ACF) to validate intrinsic periodicities and Variance Inflation Factor (VIF) analysis to eliminate multicollinearity. By pruning highly collinear features (e.g., dew_point with VIF > 200) and selecting optimal predictors—temperature, humidity, wind speed, wind direction, and MSL pressure—we ensure physical consistency and training robustness. Experimental results demonstrate that TaXer significantly outperforms current state-of-the-art models, including iTransformer [6] and TimeXer, particularly in scenarios where complex seasonal patterns are prevalent.

2. Related Work

Deep learning has revolutionized Time-Series Forecasting (TSF), transitioning from traditional statistical methods like ARIMA to sophisticated neural architectures. While Recurrent Neural Networks (RNNs) [9] and LSTMs [2] initially dominated the field, they often suffered from gradient vanishing problems in long sequences. Recently, Transformer-based models have become the de facto standard for Long-Term Time-Series Forecasting (LTSF) due to their ability to capture long-range dependencies via self-attention mechanisms. However, effectively modeling both the global periodic patterns and local temporal nuances in strongly periodic data remains a critical challenge.

2.1. iTransformer

A significant milestone in adaptation of Transformers for TSF is iTransformer [6]. Unlike standard Transformers that embed temporal tokens across variates, iTransformer inverts this structure by embedding the entire time series of each variate as a single token. This variate-based embedding strategy allows the model to learn multivariate correlations effectively via the attention mechanism. While iTransformer excels in capturing inter-variate dependencies, it primarily relies on the global view of the series, often overlooking fine-grained local temporal variations which are essential for meteorological forecasting.

2.2. TimeXer

TimeXer [13] represents the current state-of-the-art in LTSF, designed to empower Transformers with exogenous variables and advanced temporal alignment techniques. It addresses the limitations of endogenous-only forecasting by effectively incorporating external series information. As a robust baseline for our study, TimeXer demonstrates superior performance in general forecasting tasks. However, its generalized approach may not fully exploit the specific characteristics of strongly periodic data, such as the distinct seasonal and diurnal cycles found in weather datasets.

2.3. TimesNet

To explicitly model temporal variations, TimesNet [14] transforms 1D time-series into 2D tensors based on multiple periods extracted via Fast Fourier Transform (FFT). By applying 2D convolutions (Inception blocks [10]) to these transformed tensors, TimesNet effectively disentangles intraperiod- and interperiod-variations. This multi-periodicity analysis is crucial for capturing the repetitive nature of meteorological data. In our proposed TaXer, we adopt TimesNet as a core block to handle global periodic trends, complementing it with mechanisms for local feature extraction.

2.4. WaveNet

Originally proposed for raw audio generation, WaveNet [11] utilizes dilated causal convolutions to exponentially increase the receptive field without distinct loss of resolution. This architecture is exceptionally adept at capturing local dependencies and high-frequency variations in sequential data. In the context of WaXer, alternative model of TaXer, the WaveNet block serves as a specialized module for modeling local temporal contexts. By integrating WaveNet, our model ensures that rapid fluctuations in weather parameters (e.g., sudden wind gusts) are accurately forecasted alongside long-term trends.

2.5. Real-World Meteorological Dataset

While numerous benchmarks exist for TSF, high-quality datasets that reflect the complexities of real-world meteorological conditions are scarce. Existing datasets often suffer

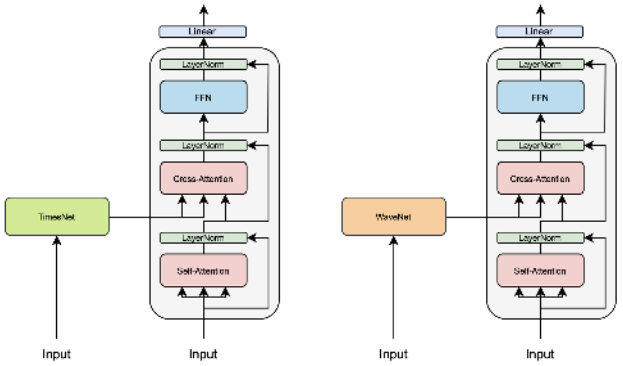


Figure 2. **Network Architecture of TaXer and WaXer.** Extracted feautres form Times-Block or Wave-Block serve as keys and values in the cross-attention mechanism.

from missing values, lack of rigorous feature selection, or weak periodicity.

3. Method

This section details our novel framework designed for effective long-term forecasting of strongly periodic sequences. Furthermore, we introduce a new benchmark dataset curated to rigorously evaluate model performance in this domain.

3.1. Feature Extraction Blocks

The proposed architecture integrates a specialized periodic feature extractor Block with a inverted Transformer backbone. We employ two distinct mechanisms to capture temporal dependencies:

Times-Block. Inspired by TimesNet, this block transforms 1D time-series into 2D tensors based on dominant periods identified via Fast Fourier Transform (FFT). It applies Inception-style 2D convolutions to capture intra-period and inter-period variations.

Wave-Block. Inspired by WaveNet, this block use dilated causal convolutions to capture multi-frequential periodic patterns. The dilation factor doubles exponentially at each layer, allowing the model to expand its receptive field without losing temporal resolution, ensuring the preservation of causality.

3.2. Network Architecture

We propose TaXer and WaXer, the hybrid forecasting models that empower the inverted Transformer framework with explicit periodic feature extraction. The overall architecture consists of three key stages as shown in Fig. 2:

1) Periodic Feature Extraction: Unlike standard Transformers that ingest raw endogenous series as an input series, TaXer first processes the input series through a Times-Block. For WaXer, this stage is replaced by a Wave-Block.

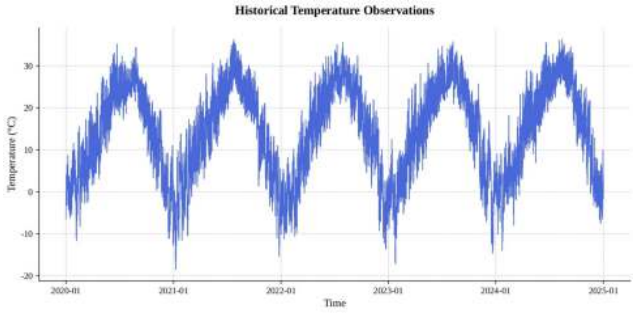


Figure 3. **Visualization of *MeteoPolitan* dataset.** The graph illustrates Time-series Variation of Hourly Temperature in Seoul.

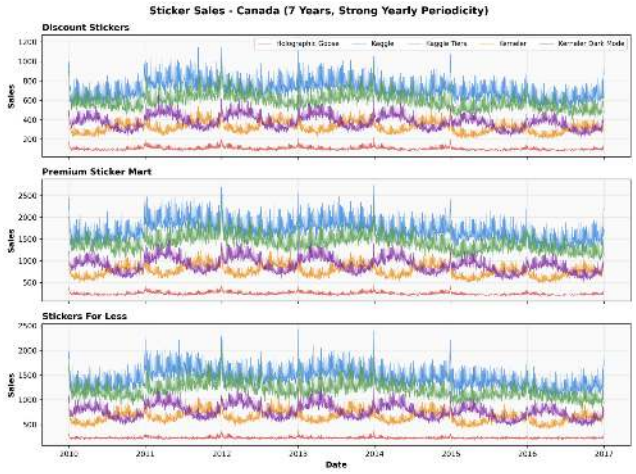


Figure 4. **Visualization of *Sticker Sales* dataset.** The graph illustrates the daily sales volume of 5 different sticker products across 3 retail stores in Canada over a 7-year period.

2) Feature Projection: The extracted features from Times-Block (or Wave-Block) are projected via a linear layer to align with the embedding dimension (d_{model}) of the forecasting backbone. These projected features serve as the "Global Exogenous Tokens."

3) Exogenous-to-Endogenous Cross-Attention: The core forecasting module utilizes an inverted Transformer architecture. The endogenous input (target series) is segmented into non-overlapping patches to form queries. The extracted global exogenous tokens serve as keys and values in the cross-attention mechanism. This design allows the model to dynamically attend to global periodic structures while predicting the future horizon.

3.3. Benchmark Datasets

To rigorously evaluate model performance on strongly periodic data, we propose a new benchmark dataset, *MeteoPolitan*, alongside the established *Sticker Sales* dataset. This combination provides a diverse evaluation environment ranging from meteorological physics to commercial

trends. Visualization of each datasets are shown in Fig. 3 and Fig. 4.

***MeteoPolitan* (Global & Korean).** We constructed a high-fidelity weather dataset by collecting hourly observation records over a 5-year period (2020–2024) from global authorities, including the KMA [5], NOAA [8], JMA [3], and DWD [1]. The dataset is subdivided into Global (5 major cities: Berlin, LA, NYC, Seoul, Tokyo) and Korean (5 major cities: Busan, Daegu, Gangneung, Gwangju, Seoul) subsets. To ensure the dataset's relevance for periodic forecasting, we employed ACF analysis to validate intrinsic periodicities and Variance Inflation Factor (VIF) analysis to eliminate multicollinearity. By pruning highly collinear features (e.g., `dew_point` with VIF $>$ 200), we identified five key variables exhibiting the highest periodicity: Temperature, Humidity, Wind Speed, Wind Direction, and MSL pressure. All data were rigorously aligned into a unified format to ensure consistency across different regions, with linear interpolation applied to handle missing values and guarantee data continuity.

***Sticker Sales* [4].** We also incorporate the *Sticker Sales* dataset, a retail benchmark characterized by strong yearly and weekly seasonality. This dataset spans 7 years of daily sales records across six countries (Canada, Finland, Italy, Kenya, Norway, and Singapore). It comprises a total of 90 distinct time series, constructed from a combination of 3 stores per country and 5 product categories per store ($6 \times 3 \times 5 = 90$). The data allows for the evaluation of model robustness in capturing long-term commercial trends amidst periodic fluctuations.

4. Experiments

4.1. Experimental Setup

4.1.1. Baselines

To evaluate the effectiveness of our proposed framework, we compare it against recent state-of-the-art Transformer-based models designed for long-term time series forecasting. We specifically selected models that address the limitations of canonical Transformers in capturing multivariate correlations and temporal variations.

iTransformer effectively captures multivariate correlations and has demonstrated superior performance over previous baselines like Informer and Autoformer. We include this to benchmark our model against the leading multivariate modeling approach.

TimeXer excels in handling complex temporal patterns and external dependencies. Since our work focuses on strongly periodic sequences, comparing against TimeXer provides a rigorous assessment of our model's ability to model temporal variations relative to the latest periodic-aware architecture.

4.1.2. Evaluation Metrics

We employ a comprehensive set of five evaluation metrics to assess model performance from multiple perspectives, including prediction accuracy, scale invariance, and temporal correlation.

Error-based Metrics (MSE, MAE, RMSE). We use Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) to penalize large prediction deviations, which is crucial for safety-critical forecasting. Mean Absolute Error (MAE) is utilized to measure the average magnitude of errors without considering their direction, providing a robust assessment of general accuracy.

Relative Metric (MAPE). Mean Absolute Percentage Error (MAPE) is employed to evaluate performance across datasets with varying scales. This ensures that our evaluation remains consistent regardless of the absolute magnitude of the target values.

Correlation Metric (CORR). Given our focus on strongly periodic sequences, capturing the phase and shape of the waveform is as important as value accuracy. We use the Empirical Correlation Coefficient (CORR) to measure the temporal alignment between the predicted sequence and the ground truth. A higher CORR indicates the model's superior capability in preserving intrinsic periodic patterns.

4.1.3. Implementation Details

Environment and Hardware. All experiments were implemented using PyTorch 2.8.0 on a Linux server running Ubuntu 20.04. The hardware infrastructure consists of dual AMD EPYC 7763 64-Core CPUs and eight NVIDIA RTX A5000 GPUs. The environment is accelerated with CUDA 12.2 and supported by key scientific computing libraries, including NumPy 1.26.4, Pandas 2.3.3, and Scikit-learn 1.6.1.

Hyperparameter Settings. We trained our model for 128 epochs with a batch size of 32. The optimization was performed using the AdamW optimizer [7] with an initial learning rate of 1×10^{-4} and momentum parameters $\beta_1 = 0.9, \beta_2 = 0.99$. For the model architecture, we set the input lookback window (L) to 512 and the single forward pass prediction horizon (T) to 16. The Transformer backbone is configured with 2 encoder layers (e_{layers}), 4 attention heads (n_{heads}), a model dimension (d_{model}) of 64, and a feed-forward network dimension (d_{ff}) of 128.

4.2. Experiment Results

Table. 1 summarizes the quantitative comparison of our proposed frameworks (TaXer and WaXer) against state-of-the-art baselines. The results clearly demonstrate that TaXer achieves consistent superiority across most datasets and metrics.

4.2.1. Quantitative Analysis

Robustness in Strongly Periodic Domains. As hypothesized, our model exhibits a significant performance gap in

Model	Dataset	MSE	MAE	RMSE	MAPE	CORR
iTransformer	Global	0.6491	0.7387	0.8595	2.8568	0.4564
	Korean	0.5796	0.6093	0.7806	2.9365	0.5143
	Sticker	0.2495	0.3715	0.4995	1.7889	0.7368
TimeXer	Global	0.6255	0.6614	0.8132	**2.0352**	0.4944
	Korean	0.5505	0.5290	0.7273	2.3945	0.5544
	Sticker	0.1647	0.3077	0.4058	**1.1816**	0.8152
WaXer (Ours)	Global	0.6253	0.6624	0.8139	2.0838	0.4882
	Korean	0.5515	0.5312	0.7288	2.4390	0.5504
	Sticker	0.1609	0.3052	0.4012	1.1923	**0.8239**
TaXer (Ours)	Global	**0.6135**	**0.6474**	**0.8046**	2.3100	**0.5202**
	Korean	**0.5413**	**0.5221**	**0.7226**	**2.3204**	**0.5587**
	Sticker	**0.1569**	**0.2971**	**0.3961**	1.1868	0.8196

Table 1. **Performance comparison of different models across Global Weather, Korean Weather, and *Sticker Sales* datasets.** The best results are highlighted in **bold**.

the Global and Korean weather datasets of *MeteoPolitan*, which are characterized by distinct physical periodicities. Specifically, TaXer outperforms iTransformer and TimeXer in MSE and MAE by a substantial margin. This confirms that explicitly modeling the periodic structure allows for more precise forecasting than merely relying on implicit attention mechanisms.

Temporal Alignment Capability. A key observation is the superiority of our methods in the CORR metric. The consistently higher CORR scores of TaXer indicate that our model not only minimizes numerical deviation but also accurately captures the phase and trend of the underlying periodic signals. This is critical for applications where preserving the temporal shape is as important as point-wise accuracy.

4.2.2. Qualitative Analysis

To provide a qualitative assessment beyond numerical metrics, we visualize the forecasting results of TaXer compared to the baselines. Figure 5 illustrates the ground truth versus the predicted sequences for a representative sample from the *MeteoPolitan* dataset.

Capturing Peak and Phase. As shown in the visualization, existing algorithms often fail to capture rapid fluctuations or exhibit a phase lag in strongly periodic regions. In contrast, TaXer demonstrates remarkable alignment with the ground truth, accurately reconstructing both the amplitude of peaks and the temporal phase of the waveforms. Even in the *Sticker Sales* dataset, which contains high-frequency noise overlaid on seasonal trends, our model effectively filters out irrelevant noise while preserving the dominant periodic structure. This qualitative evidence reinforces the quantitative findings in Table 1, proving that TaXer is not just memorizing mean values but effectively learning the intrinsic generative mechanism of periodic time series.

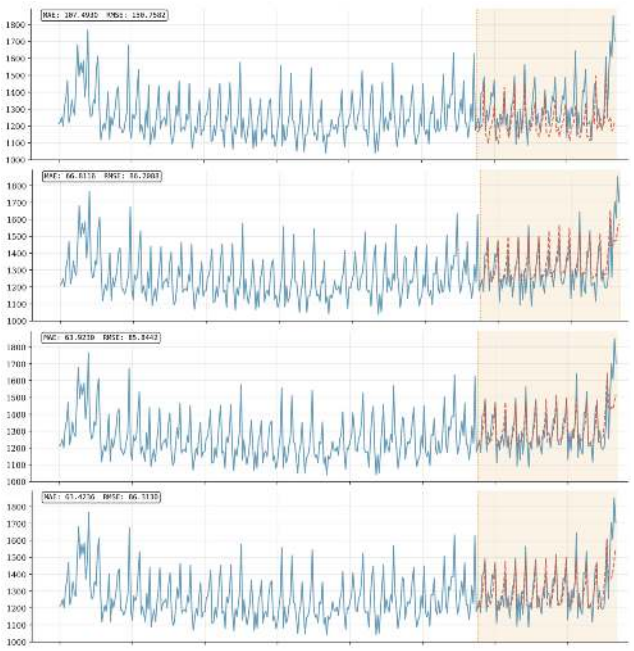


Figure 5. **Visualization of experiment results.** Qualitative visualization of temperature forecasting results on the *MeteoPolitan* (Global) dataset. The panels display the performance of iTransformer, TimeXer, WaXer, and TaXer in descending order. The blue line represents the Ground Truth, while the orange line indicates the model's prediction.

Model	Version	MSE	MAE	RMSE	MAPE	CORR
iTransformer	v1 (Baseline)	0.2870	0.4028	0.5357	1.8388	0.6870
	v2 (Small)	0.2127	0.3371	0.4612	1.5902	0.7654
	v3 (Large)	0.2563	0.3764	0.5063	1.7764	0.7264
TimeXer	v1 (Baseline)	0.1720	0.3163	0.4148	1.2582	0.8107
	v2 (Small)	0.1783	0.3181	0.4222	1.3623	0.7935
	v3 (Large)	0.1658	0.3114	0.4072	1.1607	0.8180
WaveNet	v1 (Baseline)	0.1817	0.3095	0.4263	1.5762	0.7860
	v2 (Small)	0.2314	0.3617	0.4810	1.5902	0.7317
	v3 (Large)	0.1791	0.3075	0.4232	1.5887	0.7938
TimesNet	v1 (Baseline)	0.1814	0.3121	0.4260	1.5241	0.7943
	v2 (Small)	0.1796	0.3096	0.4238	1.5933	0.7933
	v3 (Large)	0.1765	0.3111	0.4202	1.5044	0.7973
WaXer	v1 (Baseline)	0.1707	0.3150	0.4132	1.2378	0.8125
	v2 (Small)	0.1763	0.3173	0.4199	1.3298	0.7973
	v3 (Large)	0.1644	0.3081	0.4055	1.2299	**0.8213**
TaXer	v1 (Baseline)	0.1569	0.2971	0.3961	1.1868	0.8196
	v2 (Small)	0.1574	0.2991	0.3968	**1.1540**	0.8169
	v3 (Large)	**0.1543**	**0.2949**	**0.3928**	1.1657	**0.8213**

Table 2. **Ablation Study: Performance comparison across different model scales.** We evaluated three versions (v1: Baseline, v2: Small, v3: Large) for each model. The best result within each model family is underlined, and the best result across all models is highlighted in **bold**.

4.3. Ablation Study

To investigate the scalability and robustness of our proposed framework, we conducted a comprehensive ablation study by varying the model capacity. We defined three configurations for all models: v1 (Baseline), v2 (Small) with half reduced parameters, and v3 (Large) with twice increased parameters. Table. 2 summarizes the performance comparison across these configurations on the *Sticker dataset*.

As presented in Table. 2, TaXer consistently outperforms all baseline models across every model size configuration. A critical observation is the positive correlation between model capacity and performance in our framework. Also while TaXer v3 (Large) achieves the best performance with an MSE of 0.1543 and CORR of 0.8213, the standard iTransformer exhibits performance degradation when scaled up (MSE increases from 0.2127 in v2 to 0.2563 in v3). This suggests that canonical Transformers are prone to overfitting on strongly periodic datasets, whereas TaXer effectively leverages the additional capacity to capture complex periodic details without suffering from optimization difficulties. It is also noteworthy that even the lightweight version, TaXer v2 (Small), demonstrates competitive performance (MSE 0.1574), surpassing the best checkpoints of TimeXer (MSE 0.1658). This indicates that the superiority of TaXer stems not merely from parameter size but from the structural advantage of explicitly integrating periodic features with the inverted Transformer backbone. Furthermore, the fact that both TaXer and WaXer surpass their respective standalone components (TimesNet and WaveNet) as well as the state-of-the-art TimeXer confirms the significant efficacy of our synergistic hybrid framework.

5. Conclusion

In this paper, we propose TaXer, a novel framework for long-term time series forecasting of strongly periodic sequence. Our architecture, which integrates dedicated feature extraction block —Times-Block from TimesNet— before the cross-attention of inverted Transformer, demonstrated superior efficacy in capturing complex temporal patterns. Furthermore, for rigorous evaluation of periodicity driven long-term forecasting models in real-world, we introduce *MeteoPolitan*, a high-fidelity benchmark dataset meticulously curated from global and Korean meteorological records. By establishing this new standard, we aim to facilitate sensitive evaluation, addressing the limitations of existing public benchmarks. Extensive experiments on three datasets—*MeteoPolitan* (Global, Korean) and *Sticker Sales*—confirmed that TaXer consistently outperform existing state-of-the-art models, including iTransformer and TimeXer. In addition, WaXer emerged as a highly efficient alternative, offering inference speeds over two times faster than TaXer while showing slightly decreased accuracy. This highlights a distinct trade-off between precision and efficiency within our framework, allowing users to select the most suitable architecture based on their specific computational constraints.

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